Prob.

2 In the derivation of the partial trace last week using adjoints and preservation of probability, we showed in general that,

i) Given the linear map:

$$\boxed{\begin{array}{l} T : L(H_A) \longrightarrow L(H_A \otimes H_B). \\ \quad M \longmapsto M \otimes I_B \end{array}}$$

ii) It's adjont, $T^\dagger$ is characterized via inner trace product by:

$$tr\left(T^\dagger(X)\, M\right) = tr\left(X\, T(M)\right)$$

iii) And the adjoint is precisely the "throw away B" operator we are looking for:

$$T^\dagger(X) = tr_B X$$
$$= X_A \in L(H_A).$$

iv.) So in general,

$$\boxed{Tr\,(X_A M) = tr\,(X\,(M \otimes I_B)).}$$

→ this holds in , any $X \in L(H_A) \otimes L(H_A)$
$M \in L(H_A)$
$X_A = tr_B X \in L(H_A).$

v) for quatum measurements & their probabilities we specialize this to:

- $X \longrightarrow \rho_{AB}$
- $X_A \longrightarrow \rho_A = tr_B\, \rho_{AB}$
- $M \longrightarrow P_m$ (projector)
  E_m (POVM ⟶ too come!).
  M (a proper Hermitian observable.).

<u>Prob. 2</u>: Show this explicitly for $M = P_m$ a projector and $X = \rho_{AB}$: That,

$$p(m) = \mathrm{tr}\, \rho_A P_m = \mathrm{tr}\left(\rho_{AB}\left(P_m \otimes I\right)\right) \qquad (*)$$

where $\rho_A = \mathrm{tr}_B\, \rho_{AB}$.

- start with the explicit partial trace of ρ_{AB}:

$$\rho_A = \sum_j \left(I \otimes \langle j|\right) \rho_{AB} \left(I \otimes |j\rangle\right)$$

- then subs. into $(*)$ and check lhs = rhs.

LHS: $\mathrm{tr}\, \rho_A P_m$

$$= \mathrm{tr} \sum_j \left(I \otimes \langle j|\right) \rho_{AB} \left(I \otimes |j\rangle\right) P_m$$

$$= \sum_j \mathrm{tr}\left(I \otimes \langle j|\right) \rho_{AB} \left(I \otimes |j\rangle\right) |m\rangle\langle m|$$

$$= \sum_j \mathrm{tr}\, \left(I \otimes \langle j|\right) \rho_{AB} \left(|m\rangle \otimes |j\rangle\right) \langle m|$$

$$= \sum_j \mathrm{tr}\, \langle m| \left(I \otimes \langle j|\right) \rho_{AB} |mj\rangle$$

$$= \sum_j \mathrm{tr}\left(\left(\langle m| \otimes \langle j|\right) \rho_{AB} |mj\rangle\right).$$

$$= \sum_j \mathrm{tr}\, \langle mj| \rho_{AB} |mj\rangle$$ ← note: trace of scalar is just the scalar.

$$\boxed{\mathrm{LHS} = \sum_j \langle mj| \rho_{AB} |mj\rangle}$$

RHS :

$tr\, \rho_{AB} (P_m \otimes I)$.

$= tr\, \rho_{AB} (|m\rangle\langle m| \otimes I)$.

$= \sum_{ij} \langle ij| \rho_{AB} (|m\rangle\langle m| \otimes I)\, |ij\rangle$.

$= \sum_{ij} \langle ij| \rho_{AB} \left(|m\rangle\langle m|i\rangle \otimes |j\rangle \right)$.

$= \sum_{ij} \delta_{mi} \langle ij| \rho_{AB} |mj\rangle$

$\boxed{RHS = \sum_{j} \langle mj| \rho_{AB} |mj\rangle} \Longrightarrow$ LHS = RHS .

$\therefore$ holds for $M = P_m$.

- we will generalize soon from projective measurements to POVM's.

- in-stead we use a set of operators $\{E_m\}$ s.t.,

  ① $E_m \geq 0$

  ② $\sum_m E_m = I$

$\Rightarrow$ will be seen to apply to more general situations.

To Come Soon!

Prob. 3.)

We show non-selective measurement commutes with the partial trace.

- let $\{P_m\}$ be a set of projection measurements on system A
- usual measurement:

$$\rho \longrightarrow \boxed{M} \xrightarrow{\text{measured } (m)} \rho'(m) = \frac{P_m \rho P_m}{p(m)}$$

→ measurement outcome of eigenvalue m of obsv'l M.

where $p(m) = \mathrm{tr}\, P_m \rho$

- non-selective measurement.

→ do the measurement but ignore the outcome.

$$\rho' = \sum_m \rho'(m)\, p(m).$$ (average outcomes).

$$\rightarrow \boxed{\rho' = \sum_m P_m \rho P_m}$$

- Show: If we extend the non-sel've meas. on A to a system AB, such that:

$$\rho'_{AB} = \sum_m (P_m \otimes I_B)\, \rho_{AB}\, (P_m \otimes I_B)$$

and we trace out B, we get the same result on A,

$$\rho'_A = \sum_m P_m \rho_A P_m \qquad \text{where } \rho_A = \mathrm{tr}_B\, \rho_{AB}.$$

How?

We have $\rho'_{AB} = \sum_m (P_m \otimes I_B)\, \rho_{AB}\, (P_m \otimes I_B)$ given.

We want to show:

$$\rho'_A = \boxed{tr_B\, \rho'_{AB} = \sum_m P_m\, \rho_A\, P_m}$$

where: $\rho_A = tr_B\, \rho_{AB}$

$$= \sum_b (I \otimes |b\rangle)\, \rho_{AB}\, (I \otimes |b\rangle).$$

Start $\bar{c}$ LHS:

$$tr_B\, \rho'_{AB} = tr_B \sum_m (P_m \otimes I_B)\, \rho_{AB}\, (P_m \otimes I_B)$$

$$= \sum_{mb} (I \otimes \langle b|)(P_m \otimes I_B)\, \rho_{AB}\, (P_m \otimes I_B)(I \otimes |b\rangle).$$ "push out operator"

$$= \sum_{mb} P_m\, (I \otimes \langle b|)\, \rho_{AB}\, (I \otimes |b\rangle\, P_m$$

$$= \sum_m P_m \left[\underbrace{\sum_b (I \otimes \langle b|)\, \rho_{AB}\, (I \otimes |b\rangle)}_{\rho_A = tr_B\, \rho_{AB}.} \right] P_m$$

$$= \sum_m P_m\, tr_B\, \rho_{AB}\, P_m$$

$$\boxed{tr_B\, \rho'_{AB} = \sum_m P_m\, \rho'_A\, P_m}$$

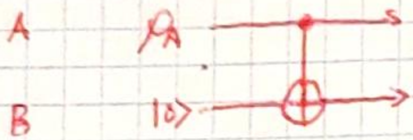
Prob. 4.)

Enviroment induced de-phasing via partial trace.

- Take a qubit A in a general state:

$$\rho_A = \begin{pmatrix} \alpha & \beta \\ \beta^* & \gamma \end{pmatrix} \qquad \alpha, \gamma \geq 0, \ \alpha + \gamma = 1.$$

- Let an enviroment B start in state $|0\rangle$.
- apply the unitary $U = CNOT_{A \to B}$ gate:



- compute $\boxed{\rho_A' = tr_B \, U(\rho_A \otimes |0\rangle\langle 0|)U^\dagger}$

Sol^{n}:

- expand matrices explicitly...

$$\rho_A \otimes |0\rangle\langle 0| = \begin{pmatrix} \alpha & \beta \\ \beta^* & \gamma \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} \alpha \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} & \beta \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \\ \beta^* \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} & \gamma \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \alpha & 0 & \beta & 0 \\ 0 & 0 & 0 & 0 \\ \beta^* & 0 & \gamma & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \equiv \begin{pmatrix} A & B \\ B^* & C \end{pmatrix}$$

$$\left.\begin{aligned} U: |00\rangle &\to |00\rangle \\ |01\rangle &\to |01\rangle \\ |10\rangle &\to |11\rangle \\ |11\rangle &\to |10\rangle \end{aligned}\right\} \quad U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} I_2 & 0 \\ 0 & S \end{bmatrix} \quad \to \text{note } U^\dagger = U \text{ here.}$$

- arguement of the trace is:

$$\begin{pmatrix} I & 0 \\ 0 & S \end{pmatrix}\begin{pmatrix} A & B \\ B^* & C \end{pmatrix}\begin{pmatrix} I & 0 \\ 0 & S \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & S \end{pmatrix}\begin{pmatrix} A & BS \\ B^* & CS \end{pmatrix} = \begin{bmatrix} A & BS \\ SB^* & SCS \end{bmatrix}$$

- so $\rho_A' = tr_B \begin{bmatrix} A & BS \\ SB^* & SCS \end{bmatrix} = tr_B \begin{pmatrix} \alpha & 0 & 0 & \beta \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \beta^* & 0 & 0 & \gamma \end{pmatrix} = \begin{pmatrix} tr\begin{pmatrix} \alpha & 0 \\ 0 & 0 \end{pmatrix} & tr\begin{pmatrix} 0 & \beta \\ 0 & 0 \end{pmatrix} \\ tr\begin{pmatrix} 0 & 0 \\ \beta^* & 0 \end{pmatrix} & tr\begin{pmatrix} 0 & 0 \\ 0 & \gamma \end{pmatrix} \end{pmatrix}$

$$\Rightarrow \boxed{\rho_A' = \begin{bmatrix} \alpha & 0 \\ 0 & \gamma \end{bmatrix}} \to \text{off-diagonals "pinched off"}$$

NOTE: $\rho_B' = tr_A(\sim)$

SAME $\Rightarrow = \begin{pmatrix} \alpha & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & \gamma \end{pmatrix} = \begin{pmatrix} \alpha & 0 \\ 0 & \gamma \end{pmatrix}$